YUHAN WEI

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Education

The University of Manchester Sep 2026 - Sep 2027 (expected)

Master of Science in Robotics

Nanjing Agricultural University Sep 2022 - Jun 2026

BEng in Electronic Information Science and Technology -- GPA 3.63

Research Experience

Development of a Behavioural Phenotyping System for Lactating Sows and Piglets Using Inspection Robots

Jan - Dec 2024

- National-level undergraduate innovation training project
- Team project on visual behaviour monitoring for lactating sows and piglets, comparing NanoDet-Plus and YOLOv8 for sow posture recognition and piglet detection.
- The system used TensorRT-accelerated deployment on Jetson Nano and supported analysis of sow behaviour patterns.

CNN-based Chrysanthemum Recognition System

Feb 2026 - Jun 2026

- Built an eight-class dataset of 1,169 images and trained an EfficientNet-B3 classifier using transfer learning; the saved model achieved 95.38% validation accuracy.
- Integrated the model into a Flask application with image upload, preview and top-three predictions with confidence scores.

Engineering Projects

Intelligent Inspection Robot for Multi-tier Caged Chicken Farms

Oct - Dec 2023

- Assisted with chicken-image collection and annotation and participated in basic model training for the team's visual monitoring system.

Agricultural Tourism and Cultural Visit System

Nov - Dec 2023

- Coordinated requirements and detailed design, integrated team code, and implemented administrator login, ODBC configuration and multimedia functions using C++ and MFC.

Intelligent Inspection Robot for Sows' Health Monitoring

Apr - Jun 2024

- Contributed to robot hardware debugging and sensing-module integration, and participated in developing a platform to display inspection and health-monitoring information.

MATLAB-based Music Player

Jun 2024

- Designed the App Designer interface and playback callbacks; implemented playlist controls, timer-driven waveform and FFT displays, delay effects and basic stereo vocal reduction.

An Edge Computing-Based Egg Collection Conveyor Belt System for Online Egg Inspection

May - Jul 2024

- Helped test and debug DeepSORT tracking and line-crossing counting; compared automated counts with manual records to assess duplicate counts, missed counts and classification errors.

Line-Following Robot for Poultry Farm Monitoring

Nov 2024 - Jan 2025

- Implemented STM32 four-sensor line following, PWM motor control and ultrasonic obstacle avoidance; adapted OpenMV code for model-chicken detection and refined route performance through hardware testing.

Technical Skills

- **Programming:** Python, C, C++, MATLAB
- **Computer Vision & Deep Learning:** PyTorch, YOLO, Image Classification, Object Detection
- **Embedded Systems:** STM32, OpenMV

Awards

- National First Prize, 5th Global Campus Artificial Intelligence Algorithm Elite Competition.
- Third Prize, Jiangsu Division, 26th China Robotics and Artificial Intelligence Competition.
- Third Prize, Jiangsu Division, 2024 RAICOM Robotics Developer Competition.
- Team Member, National-Level Undergraduate Innovation Training Project (SRT).
- Second-Class Merit Student Scholarship, Nanjing Agricultural University (awarded twice).
- Special Scholarship for Science and Technology Innovation, Nanjing Agricultural University.
- Outstanding Student Leader, College of Artificial Intelligence, Nanjing Agricultural University.